SECTION ES

Electrical System

Table of Contents

GENERAL INSTRUCTIONS..... ES-3
BATTERY..... ES-4
CLEANING THE BATTERY TERMINALS AND CABLE CONNECTIONS ES-7
ELECTRICAL SYSTEM FUSES ES-7
FUSE PANEL RELAY AND FUSE LOCATIONS ES-7
REMOVING THE BATTERY..... ES-8
BATTERY INSPECTION..... ES-8
BATTERY INSTALLATION..... ES-9
REMOVING THE HEADLAMP BULB..... ES-9
INSTALLING THE HEADLAMP BULB..... ES-9
REMOVING THE HEADLAMP LENS ASSEMBLY - AURORA MODELS ES-10
INSTALLING THE HEADLIGHT LENS ASSEMBLY - AURORA MODELS..... ES-11
REMOVING THE HEADLAMP LENS ASSEMBLY - FRONTIER MODELS ES-11
INSTALLING THE HEADLIGHT LENS ASSEMBLY - FRONTIER MODELS ES-12
REMOVING THE TAIL LIGHT..... ES-13
INSTALLING THE TAIL LIGHT..... ES-13
REMOVING/INSTALLING STARTER MOTOR - KAWASAKI 800 ..... ES-14
REMOVING/INSTALLING STARTER MOTOR - BRIGGS 600, 650 & 700..... ES-15
REMOVING/INSTALLING STARTER MOTOR - BRIGGS 850/950..... ES-16
DIGITAL GAUGE PACKAGE - FRONTIER/AURORA..... ES-17
CALIBRATING THE DIGITAL GAUGE PACKAGE - AURORA..... ES-18
CALIBRATING THE DIGITAL GAUGE PACKAGE - FRONTIER..... ES-20
REPLACING THE ALTERNATOR V-BELT - 800 KAWASAKI MODEL..... ES-22
REMOVING THE ALTERNATOR - 800 KAWASAKI MODELS..... ES-22
INSTALLING THE ALTERNATOR - 800 KAWASAKI MODELS..... ES-23
DASH PANEL SWITCHES AND GAUGE..... ES-24
SPEEDOMETER SENSOR AND WIRING ES-25
WIRING SCHEMATIC (MODELS 800, 850, 950)..... ES-26
WIRING HARNESS ADAPTER (MODEL 800) ES-27
WIRE HARNESS CONNECTORS AND COLOUR CODES
(MODELS 800,850,950) ES-28
WIRING SCHEMATIC (MODELS 650/700 ..... ES-29
WIRE HARNESS CONNECTORS AND COLOUR CODES (MODELS 650/700) ES-30
WIRING SCHEMATIC (MODELS 600)..... ES-31
WIRE HARNESS CONNECTORS AND COLOUR CODES (MODELS 600) ES-32
WIRE HARNESS SCHEMATIC (MODEL 600 EU) ES-33
WIRE HARNESS CONNECTORS AND COLOUR CODES (MODEL 600 EU) ES-34

TACH EXTENSION WIRE HARNESS ES-35
FRONITER/AURORA GAUGE PLUG PIN MAP..... ES-36

General Instructions

WARNING

Detailed information on standard workshop and safety procedures, and general servicing operations is not included in this manual, which has been prepared to assist qualified service personnel. ARGO assumes no responsibility or liability for PERSONAL INJURY or VEHICLE DAMAGE which results from any servicing procedure performed, including those instructions outlined in this manual. Before performing a servicing operation, an individual must have determined to his/her satisfaction that a personal injury or vehicle damage will not result from the servicing procedure or tools selected.

CAUTION

To prevent damage to the electrical system:

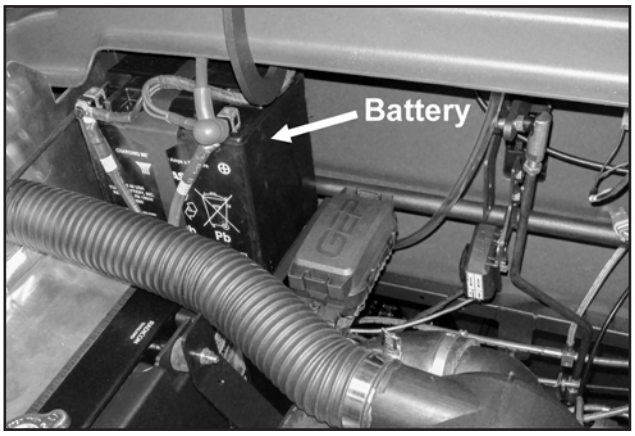
- Always disconnect the battery when servicing any electrical component.
- Never weld on the vehicle without first disconnecting both positive and negative battery cables. Make sure the part you are welding is properly grounded.
- Connecting booster cables to start your vehicle is not a recommended practice. Damage to engine electrical components could occur.
- Connect switch terminals properly, especially the ground wire.

WARNING

Battery fluid contains sulphuric acid. If battery fluid comes in contact with skin or eyes, flush thoroughly with water. If swallowed, call physician or poison control centre immediately. KEEP AWAY FROM CHILDREN. Serious personal injury can occur. Always wear rubber gloves and safety glasses when servicing the battery.

Batteries can explode and cause serious personal injury if exposed to flame or sparks. Never smoke while servicing the battery.

Battery - AGM Battery - ARGO (Yuasa)



The battery is located under the hood on the right hand side of the vehicle at the dash (intake side of vehicle)

AGM batteries look different than other battery types. They are more compact because there is no free electrolyte making them more “volume efficient” in that they take up less space for their energy storing capacity. Because they are filled with electrolyte only one time during activation there are no filler caps. Instead a sealing plug permanently covers the filler ports. Also there is no vent tube. AGM batteries are ideal for long term storage. In addition, an AGM battery is an excellent choice for use in vehicles where acid spills could occur (atv’s etc.).

Activating and Charging AGM Batteries

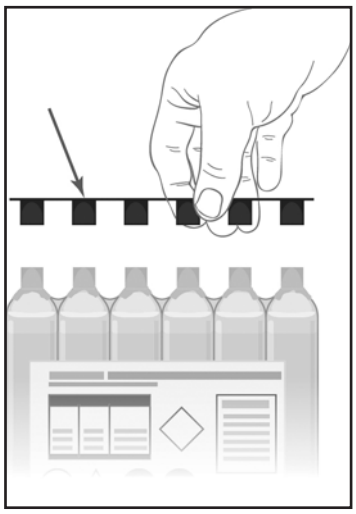
Activating an AGM battery is a simple process and differs from activating a Conventional battery. Un-activated, AGM batteries can be stored for long periods of time as long as they are kept in a cool, dry location and out of direct sunlight. Also the foil sealing strip covering the filler ports should not be removed until the battery is ready to be activated. Use only the electrolyte container that comes with the battery for filling the cells as it has a higher concentration of sulfuric acid than the acid used for Conventional batteries.

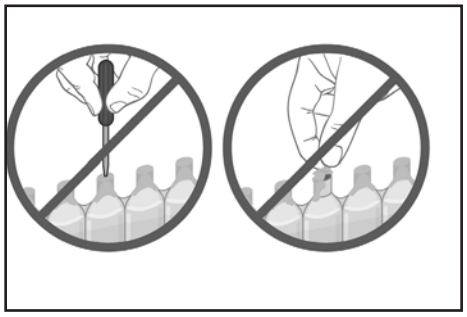


The electrolyte container that is shipped with a dry AGM battery contains the correct amount of battery acid and is more concentrated than the electrolyte used in a conventional battery. All AGM battery electrolyte containers are not the same. Each contains the proper amount of electrolyte for its specific battery.

Before filling, read the electrolyte handling instructions and precautions on the label. Do not smoke when activating a battery or handling battery acid. Always wear plastic gloves and protective eyewear and be sure to read the Battery Safety section that comes with your battery. The following seven steps should be used to activate an AGM battery:

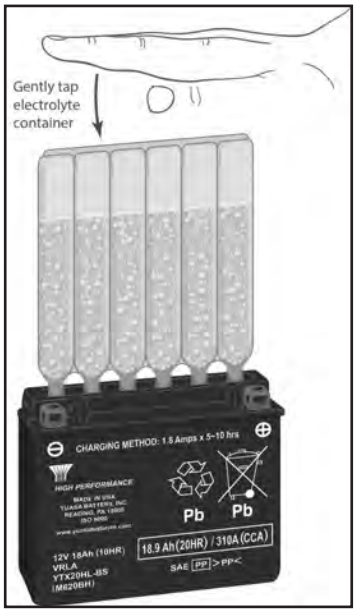
The battery must be out of the vehicle and placed on a level surface. Remove electrolyte container from the plastic storage bag. Remove the strip of caps. **Put the strip aside as you will use it later to seal the battery cells**. For battery filling use only the dedicated acid container that comes with the battery as it contains the proper amount of electrolyte for that specific battery. This is important to service life and battery performance. Do not pierce, or otherwise open the foil seals on the electrolyte container. Do not attempt to separate the individual electrolyte containers.





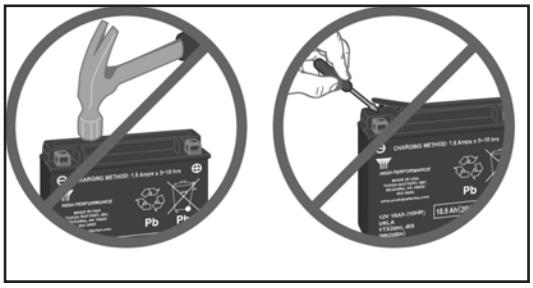
Place the electrolyte container with the foil seals facing down into the cell filler ports on the battery. Hold the container level and push down to break the foil seals. Electrolyte will start to flow into the battery and air bubbles will be seen inside the container. Do not tilt the electrolyte container.

Check the electrolyte flow. Keep the container in place for 20 minutes or longer until it empties completely. If no air bubbles are coming up from the filler ports, or if container cells haven't emptied completely after 20 minutes, tap the container and/or battery case a few times to cause the electrolyte to flow into the battery. Do not remove the acid container from the battery until it is completely empty. The battery requires all of the electrolyte from the container for proper operation.



Bubbles will appear as the electrolyte container fills the battery indicating the flow of battery acid. Tap the container periodically to keep the electrolyte flowing until the container is completely empty. Never puncture the top of the acid container to speed up the filling process.

Remove the empty electrolyte container from the battery. Fully insert the strip of sealing caps (previously removed from the electrolyte container) into the battery filling ports. Make sure the strip of caps is fully inserted and flush with the top of the battery. Insert the caps by hand, do not use a hammer or excessive force. Never remove the strip of caps or add water or electrolyte to the battery during its service life.



After filling, the sealing caps should be installed using hand pressure only. The sealing cap should never be removed once the battery is activated.

For batteries with ratings of less than 18 AH, let the battery stand for 20 to 60 minutes. For batteries with higher AH ratings, and/or having the High Performance rating (designated by an “H” in the part number/name) allow the battery to stand for 1 to 2 hours. Yuasa AGM batteries have the amp hour (AH) printed on the front of the battery case. The stand, or rest period, allows the electrolyte to permeate into the plates for optimum performance.

Newly activated AGM batteries require an initial charge. After adding electrolyte, a new battery is approximately 75-80% charged. After the “stand” period (step 6), charge the battery to bring it to a full state-of-charge. The battery charger used for initial charging should be able to charge at 12.8+-volts for an AGM battery. All Yuasa battery chargers are capable of reaching this minimum voltage and initializing/activating an AGM battery.

Points to Remember

- Newly activated AGM and Conventional batteries require an initial charge before being placed into service
- Yuasa Smart Battery Chargers use constant current and pulse technology and can activate, charge and maintain all Yuasa batteries
- High-rate, automotive types of chargers can cause damage to powersports batteries
- High performance batteries (designated by an “H” in the part number) must stand 1 to 2 hours after initial charging
- Never remove the strip of caps on an AGM battery to add water or electrolyte during its service life
- Before activation of Conventional batteries remove the red sealing cap from the vent elbow and discard it

WARNING

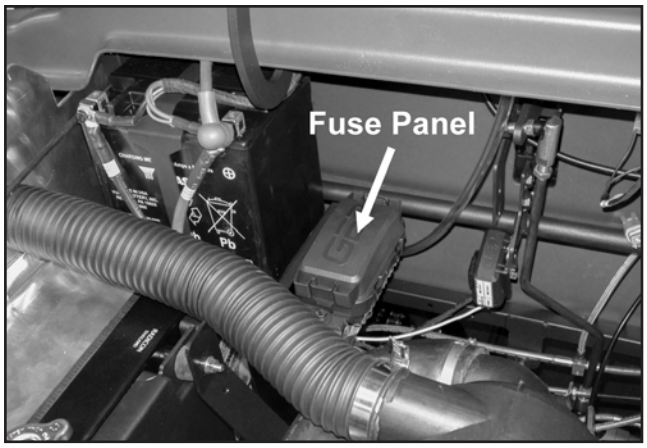
Ventilate area when charging. Keep away from spark, heat, cigarettes or open flame.

Cleaning the Battery Terminals and Cable Connections

Clean the battery terminals and cable connections every 100 hours. Remove the black NEGATIVE (-) cables first. Make sure you reconnect the NEGATIVE (-) cables to the NEGATIVE (-) post and the red POSITIVE (+) cables to the POSITIVE (+) posts. Damage to the electrical system will occur if the cables are reversed.

Electrical System Fuses

All models of the ARGO are equipped with push-in type automotive fuses. The fuses protect the electrical circuits of the vehicle. They are located in a sealed fuse block, inside the engine compartment, on the right hand side of the vehicle closest to the firewall (intake side).



Fuse Panel Relay And Fuse Locations

Aurora Models

MAIN RELAY		LIGHTS ENABLE RELAY	ACC. RELAY	OPTION - A RELAY	HOOD FAN RELAY
ECM 10 A	GAUGE 2 A	BRAKE FAN 7.5 A	HOOD FAN 20 A	OPTION - B RELAY	START ENABLE RELAY
STARTER 10 A	BILGE SINGLE 5 A DUAL 10A	USB OUTLET 5 A	HORN 10 A		
HIGH BEAM 15 A	WINCH 5 A	DASH 7.5 A	LOW BEAM RELAY	HIGH BEAM RELAY	STARTER RELAY
LOW BEAM 15 A	WIPER 15 A	CHAIN OILER 10 A			

FUSE PANEL 600-0157

Empty	Empty	Empty

Accessory Block

Ground Block

Frontier Models

ECM 10 A	12V OUTLET 5 A	MAIN RELAY		GROUND RELAY	ACCESSORIES RELAY
STARTER 10 A	BILGE PUMP 10 A				
GAUGE 2 A	BRAKE FAN 7.5 A	START ENABLE RELAY	STARTER RELAY	LOW BEAM RELAY	HIGH BEAM RELAY
DASH 7.5 A	OPTION - A				
LOW BEAM 15 A	OPTION - A	OPTION - B	OPTION - B	OPTION - A RELAY	OPTION - B RELAY
HIGH BEAM 15 A	FRONTIER 10 A	GROUND BAR			

FUSE PANEL 600-0041

NOTE

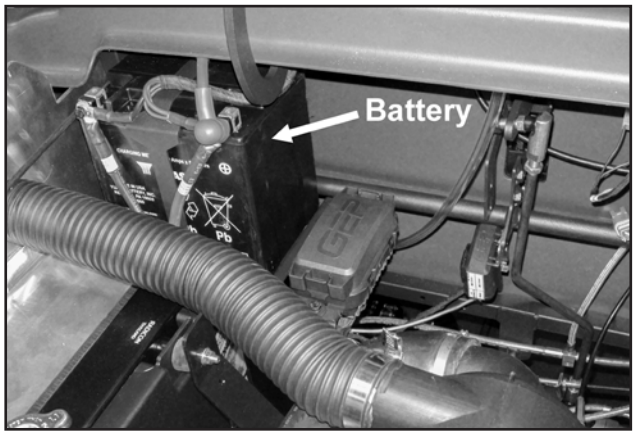
Keep the battery charged. Never allow the battery to become drained by unnecessary use of the headlights or over cranking of the engine.

If the battery is dead or the electrical charge is low, re-charge it with a 12 volt battery charger. Slow charge for at least 6 hrs.

If persistent problems with battery charge occur, before replacing the battery, ensure that the engine charging system is functioning properly. Refer to the appropriate engine service manual given in section GI of this service guide.

Battery Removal

1. The battery is located under the hood on the right hand side of the vehicle at the dash (intake side of vehicle)
2. Disconnect the battery by removing the negative cable from the ground post first.
3. Remove the retaining strap securing the battery and lift the battery from the vehicle to a clean work service.



Cleaning the Battery

Clean the top of the battery every 250 hours with a mixture of baking soda and water. Before cleaning the battery, remove it from the vehicle and make sure the pod vents are in place. Soak a cloth in the soda/water mixture and scrub the top of the battery. After the foaming has stopped, flush with clean water and dry with a clean cloth.

Battery Inspection

Inspect the battery for:

- cracks in the battery casing
- cracked or broken terminals
- exposed or corroded cells
- weak cells
- corroded terminals

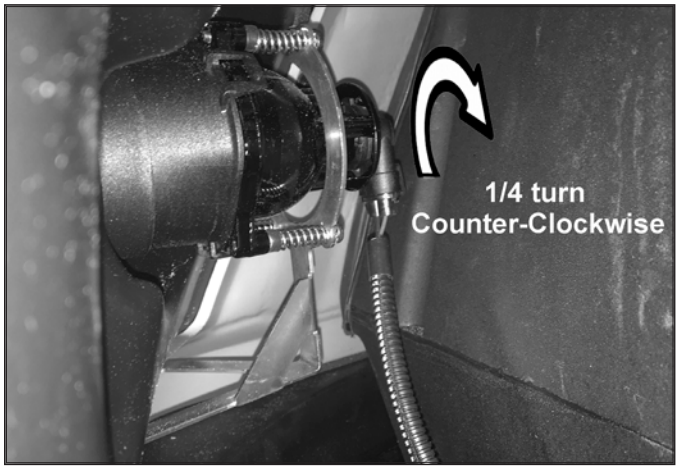
If any of these conditions are noted, replace the battery immediately.

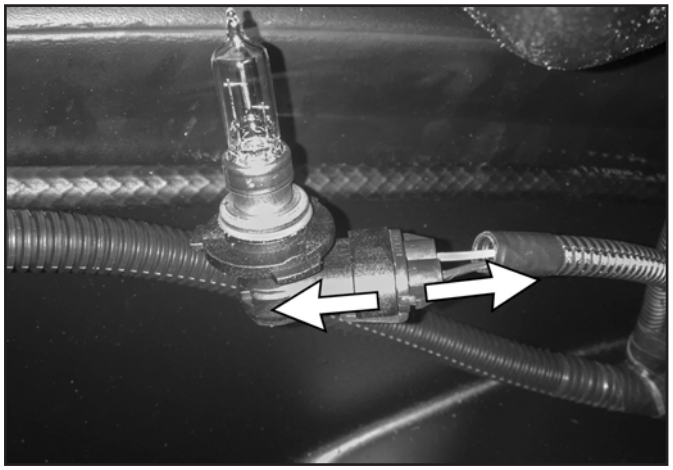
Installing the Battery

1. Place the battery back into the battery compartment. Orient the battery so the Positive Terminal (+) is to the installer's right.
2. Fasten the main power cable to the positive battery terminal first and cover with the red rubber insulator boot.
3. Re-attach the ground cable to the negative terminal side of the battery.
4. Secure with the retaining strap.
5. Reinstall the hood.

Removing the Head Lamp Bulb

1. Open the vehicle hood.
2. Locate the back of the head lamp.
3. Grasp the head lamp bulb fitting and turn 1/4 turn counter-clockwise and pull out of the lens housing. Push down on the locking tab in the connector and unplug the bulb from the wiring.





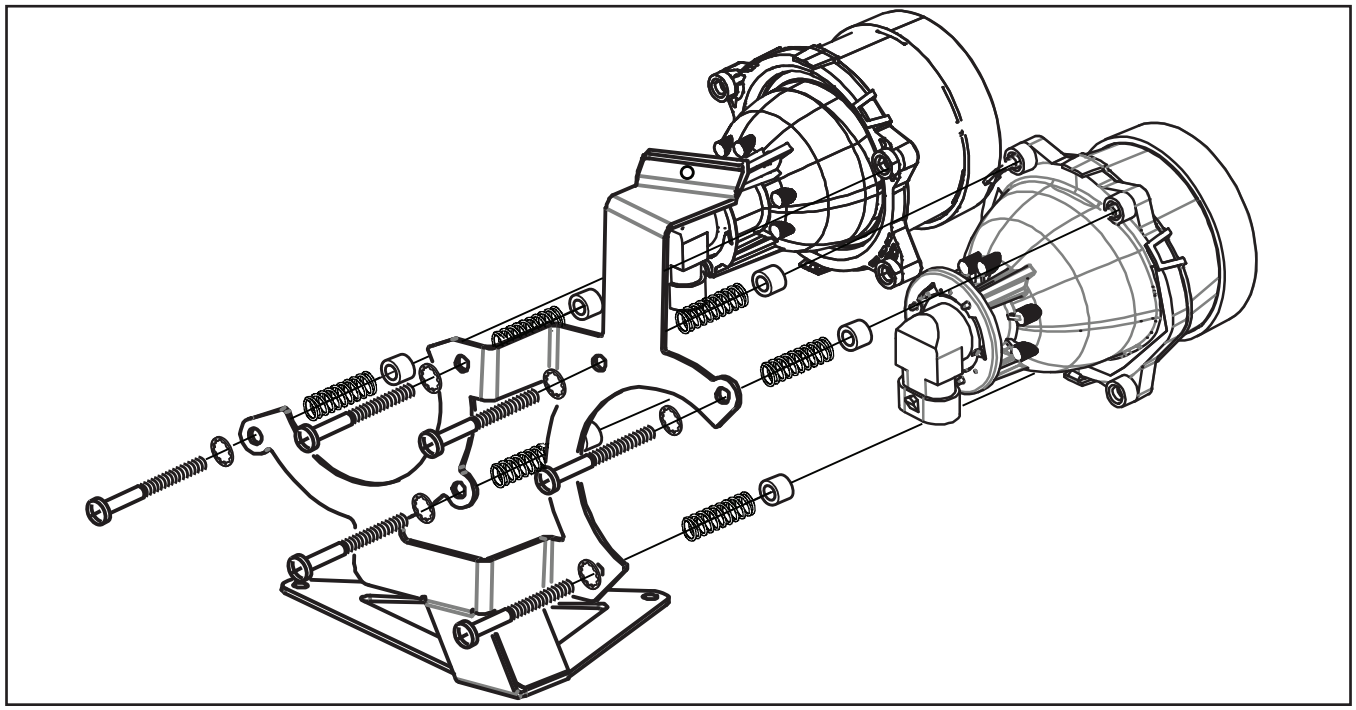
Installing The Head Lamp Bulb

CAUTION

When installing the new bulb, avoid any contamination of the glass surface. Try not to excessively touch the bulb with your fingers.

1. Clip the new bulb assembly into the wire harness.
2. Reaching in through the engine access area, align the bulb assembly with the headlamp housing. Push in and turn 1/4 turn clockwise.

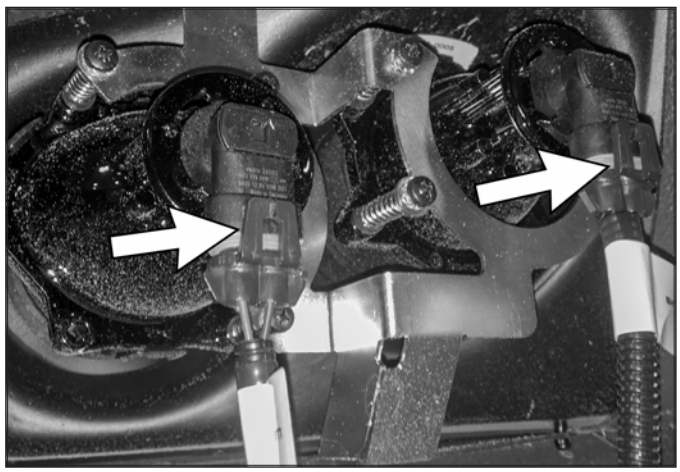
Removing the Head Lamp Lens Assembly - Aurora Models (High/Low Beam)

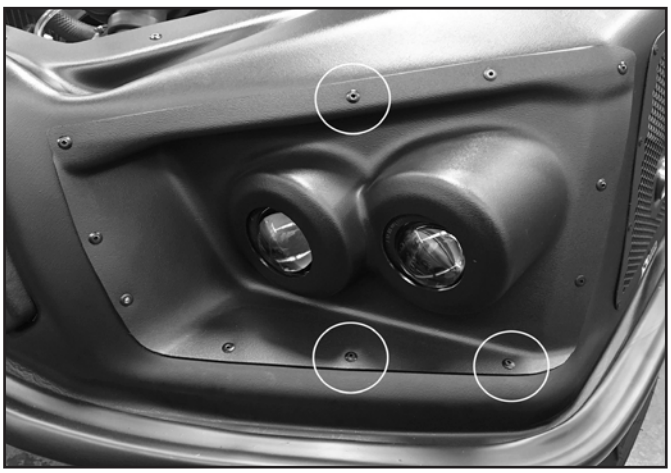


1. From the engine compartment side, disconnect the wiring from each headlamp.
2. Drill out and remove the three (3) rivets securing the headlight assembly to the front bucket.

NOTE

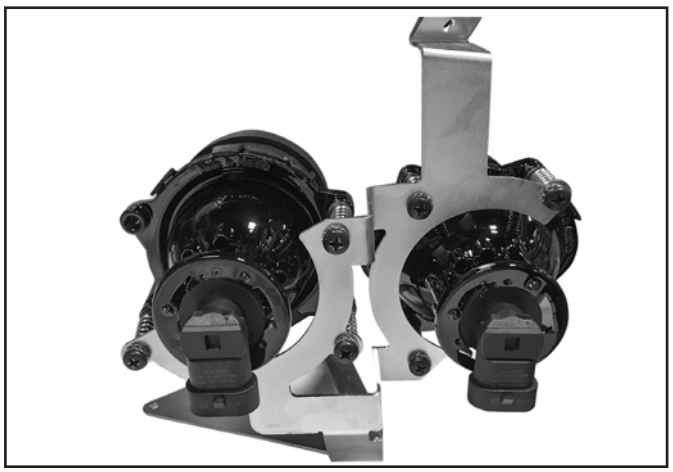
Exhaust side Light Bucket will require moving some exhaust duct shielding out of the way for access to head lamp assembly.





3. Pull the headlight lens assembly out through the hood area of the engine compartment and place on a clean work bench for disassembly.





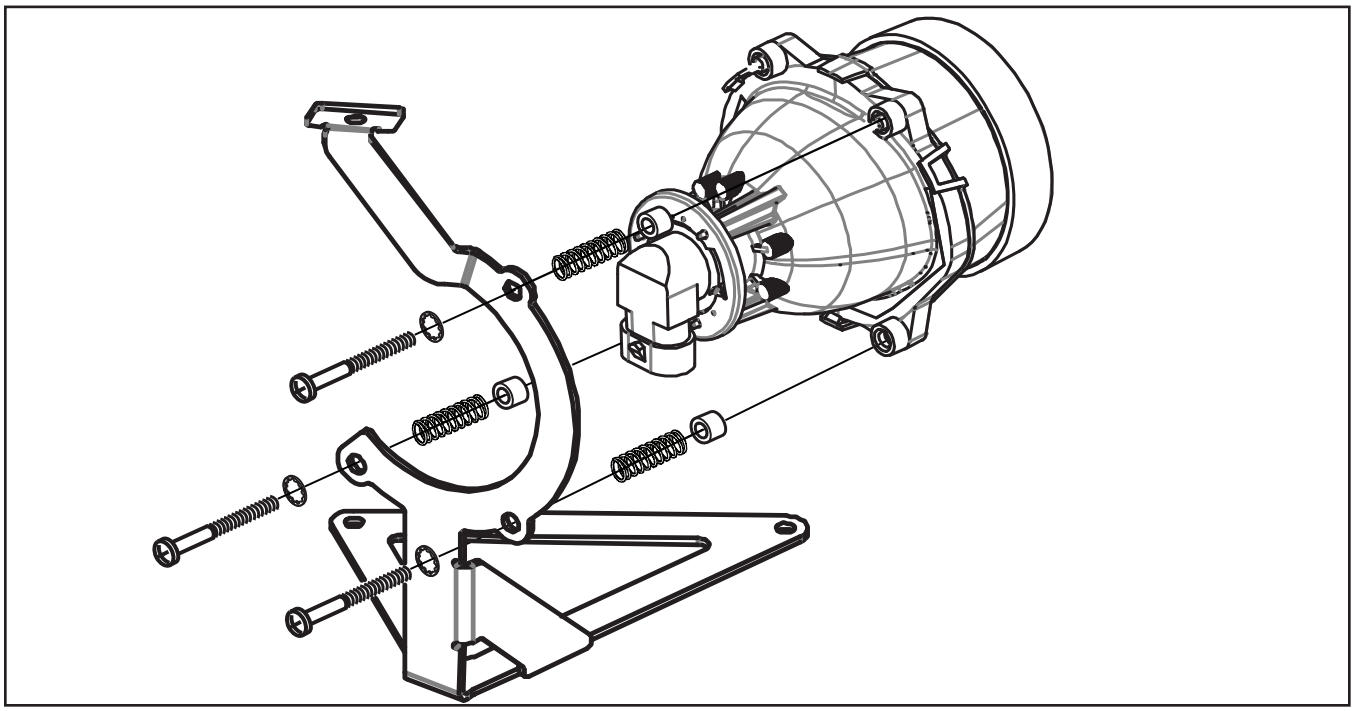
4. Replace each lens as required.

Installing The Headlight Lens Assembly

1. From the engine compartment side, maneuver the headlight/bracket assembly into position and align the holes in the mounting bracket with the 3 holes left open from the removal procedure. Secure the headlight bracket with three (3) Rivet - 3/16 x 3/4 Black

2. Reconnect head lamp wiring.

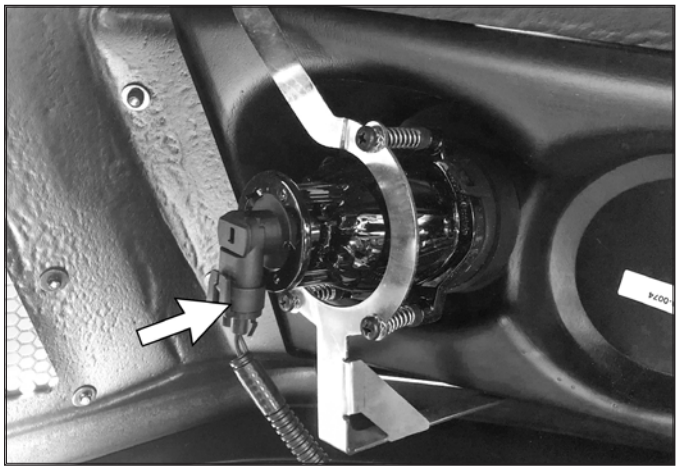
Removing the Head Lamp Lens Assembly - Frontier Models (Low Beam Only)

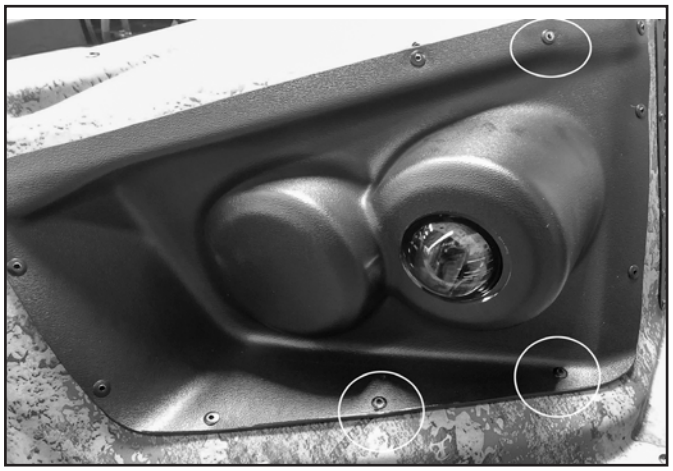


1. From the engine compartment side, disconnect the wiring from the head lamp.
2. Drill out and remove the three (3) rivets securing the headlight assembly to the front bucket.

NOTE

Exhaust side Light Bucket will require moving some exhaust duct shielding out of the way for access to head lamp assembly.





3. Pull the headlight lens assembly out through the hood area of the engine compartment and place on a clean work bench for disassembly.





4. Replace each lens as required.

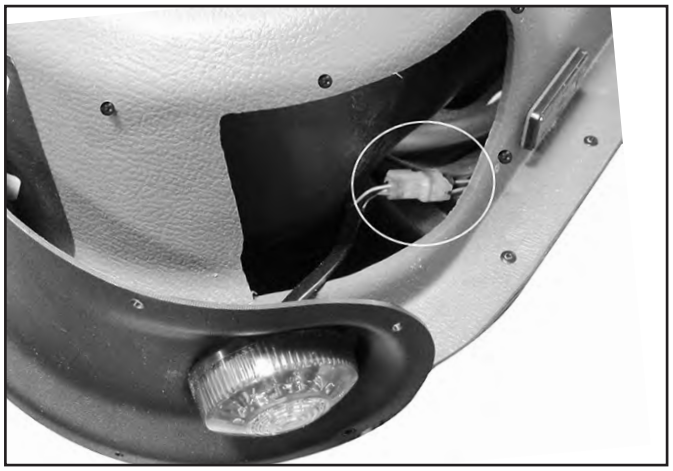
Installing The Headlight Lens Assembly

1. From the engine compartment side, maneuver the headlight/bracket assembly into position and align the holes in the mounting bracket with the 3 holes left open from the removal procedure. Secure the headlight bracket with three (3) Rivet - 3/16 x 3/4 Black
2. Reconnect head lamp wiring.

Removing The Tail Light

1. Drill out the rivets securing the rear tail light buckets to the upper body.
2. Pull the rear tail light bucket from the vehicle and unplug tail light wire connection.
3. Remove the nuts securing the tail light lens to the bucket and remove.







Installing The Rear Tail Light

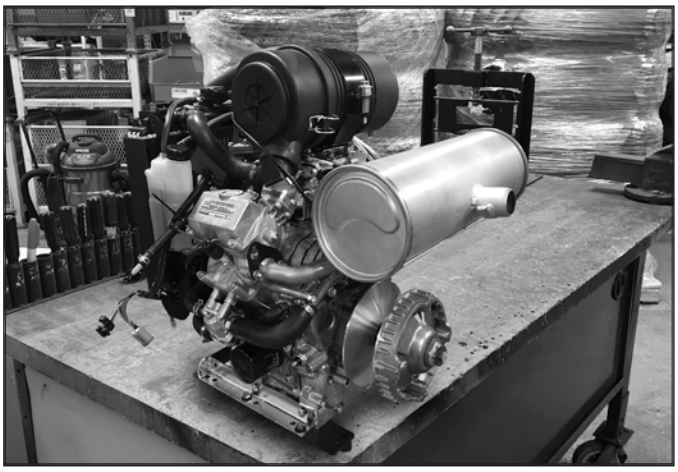
4. Replace and secure the new lens into the bucket. Plug the lens back into vehicle wire harness and reinstall the bucket to the upper body securing with the screws.

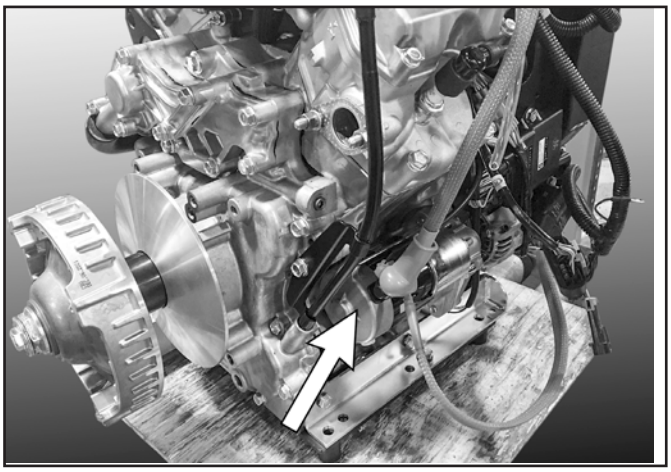
Removing Starter Motor - Kawasaki 800

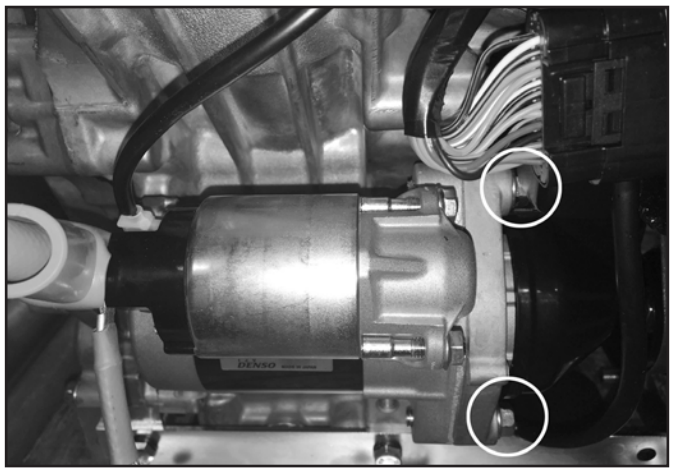
IMPORTANT

This procedure will require the removal of the engine from the vehicle for best access to the starter motor mounting fasteners. It will also ensure proper alignment during installation and proper torque values for the starter motor bolts.

1. Remove the engine from the vehicle. Refer to **Removing The Engine (Kawasaki 800)** in the engine section of this service guide.
2. Place the engine to a clean work bench.
3. Locate the starter at the rear of the engine.
4. Remove the two (2) bolts securing the starter to the engine block.
5. Slide the starter out from the blower housing side of the engine.







6. Install starter motor and torque bolts to **15 FT. LBS. (20Nm)**
7. Reinstall the engine. Refer to **Installing The Engine Briggs (600, 650 or 700)** in the engine section of this service guide.

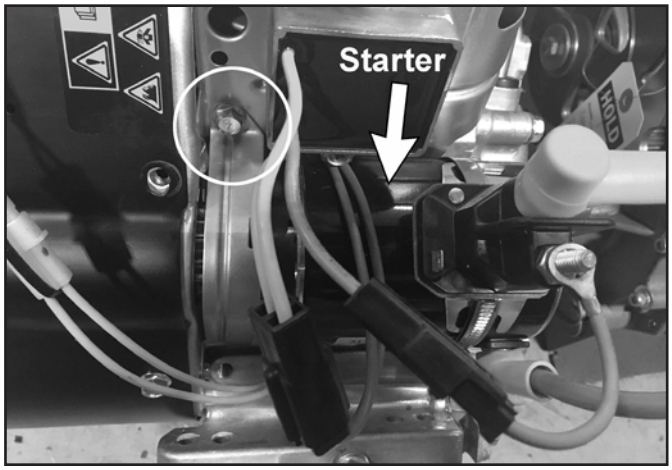
Removing Starter Motor - Briggs 600, 650 & 700

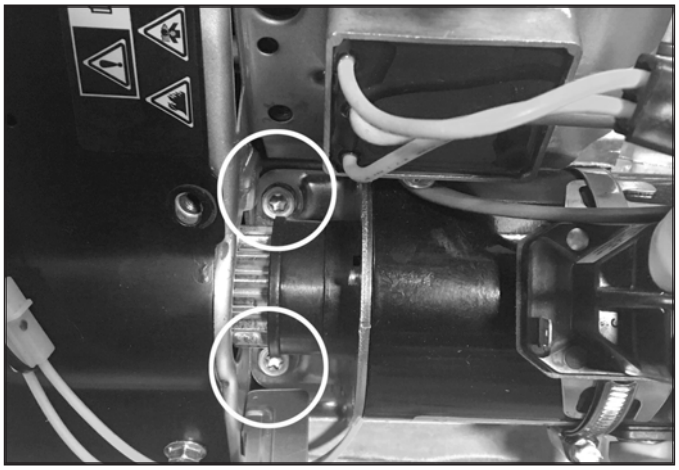
IMPORTANT

This procedure will require the removal of the engine from the vehicle for best access to the starter motor mounting fasteners. It will also ensure proper alignment during installation and proper torque values for the starter motor bolts.

1. Remove the engine from the vehicle. Refer to **Removing The Engine (Briggs 600/650/700)** in the engine section of this service guide.
2. Place the engine to a clean work bench.
3. Locate the starter at the front of the engine.
4. Remove the bolt securing the starter gear pinion guard and fold it back to expose the mounting bolts.
5. Slide the starter out from the blower housing side of the engine.







6. Install starter motor and torque bolts to **140 in. lbs. (16 Nm).** Reinstall starter gear pinion guard.
7. Reinstall the engine. Refer to **Installing The Engine Briggs (600, 650 or 700)** in the engine section of this service guide.

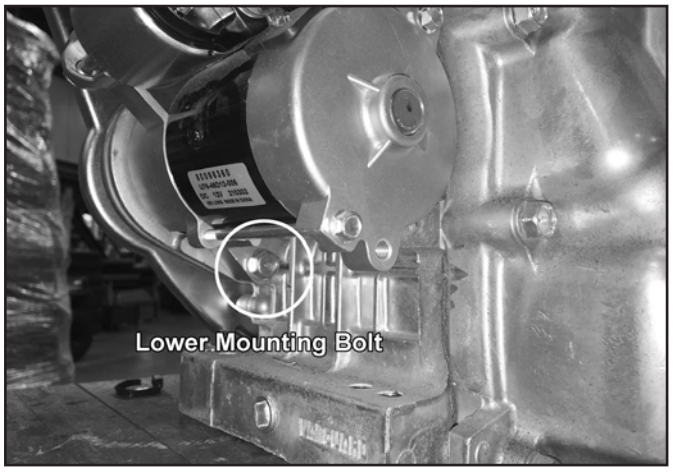
Removing Starter Motor - Briggs 850/950

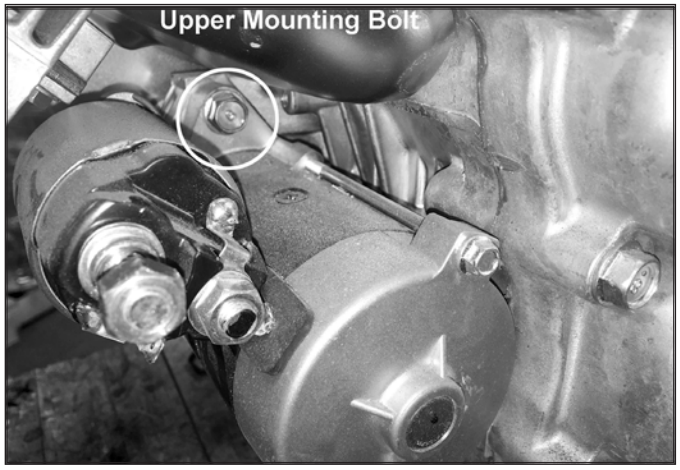
IMPORTANT

This procedure will require the removal of the engine from the vehicle for best access to the starter motor mounting fasteners. It will also ensure proper alignment during installation and proper torque values for the starter motor bolts.

1. Remove the engine from the vehicle. Refer to **Removing The Engine (Briggs 850/950)** in the engine section of this service guide.
2. Place the engine to a clean work bench.
3. Locate the starter at the front of the engine.
4. Remove the two (2) bolts securing the starter to the engine block.
5. Slide the starter out from the blower housing side of the engine.







6. Install starter motor and torque bolts to **140 in. lbs. (16 Nm)**
7. Reinstall the engine. Refer to **Installing The Engine Briggs (850/950)** in the engine section of this service guide.

Digital Gauge Packages - Aurora and Frontier

Aurora Gauge

Engine Light - Amber Light

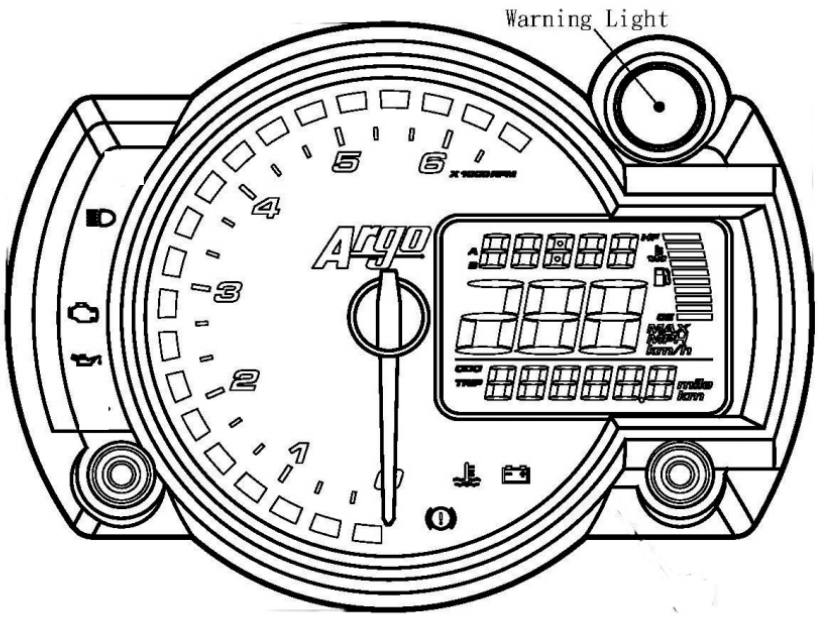
Oil Light - Red Light

Water Temperature Light - Red Light

Voltage Light - Red Light

Parking Brake Light - Red Light

Warning Light - Yellow/Amber
(Check for fault codes)

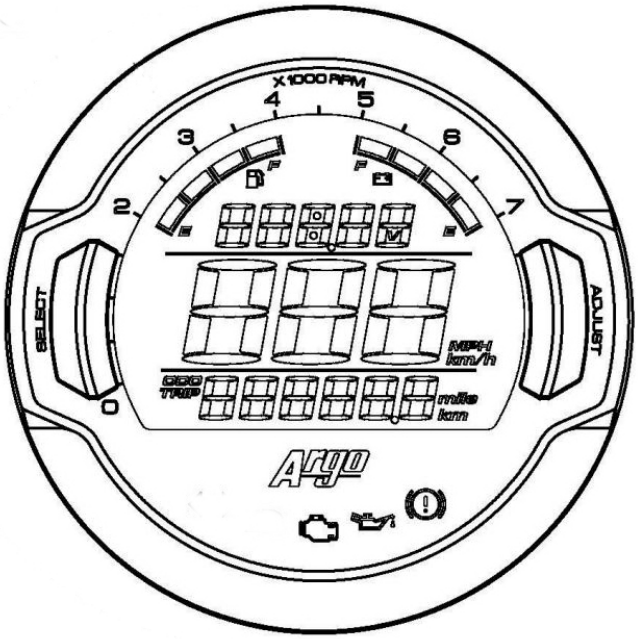


Frontier Gauge

Engine Light - Amber Light

Oil Light - Red Light

Parking Brake Light - Red Light



Calibrating the Digital Gauge Package

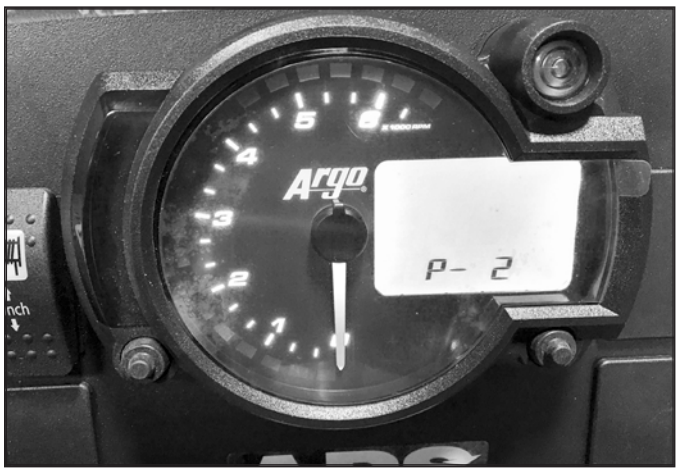
Aurora Digital Gauge Package

1. With key switch off, hold both “Select” and “Adjust” buttons down simultaneously.
2. Turn the key to the run position until you see the letter “adj” and then release both buttons.
3. Using the “Adjust” button, change the top flashing numbers to read **1994** (for 25"tires). Cycle to the next number using the “Select” button.
4. Set the lower number to **15P** for correct speedometer reading. Push **Select** button.





5. Set to P-2 for 800 Model Aurora (tach setting).
6. Set to P-1 for all models except Aurora 800 models (tach setting). Push **Select** button.





7. Set to F-1 for fuel gauge setting. Push **Select** button.